

Name _____ Surname _____ Date ____/____/____

Thermal physics · Mock midterm test · Version A

Time: 50 minutes · Total: 44 marks · Calculator, ruler and sharp pencil allowed. Section IV is in Italian: answer it in Italian or in English.

I · Multiple choice

[8]

Circle one letter for each question. Each question is worth 1 mark.

1. Which statement about heat and temperature is correct?
 - A Temperature is the amount of heat stored in a body.
 - B Heat is energy transferred because of a temperature difference.
 - C A hot body contains more heat than a cold body of the same mass.
 - D Heat and temperature are two names for the same quantity.
2. Body X is in thermal equilibrium with body Z, and body Y is also in thermal equilibrium with body Z. What follows?
 - A X must be hotter than Y.
 - B X and Y can be compared only after they have been in contact.
 - C X and Y have the same temperature, even if they never touch.
 - D Nothing can be said about X and Y.
3. A sample is at 27 °C. What is this temperature in kelvin?
 - A 300 K
 - B 246 K
 - C -246 K
 - D 27 K
4. Why is there no such thing as a temperature of -50 K?
 - A By convention, the kelvin scale was defined without negative numbers.
 - B At 0 K all particle motion stops, so lower values would have no meaning.
 - C The kelvin scale is used only for gases.
 - D 0 K is the lowest temperature possible, so nothing can be colder.
5. Two liquid-in-glass thermometers are identical, except that thermometer P has a narrower tube than thermometer Q. Which statement is correct?
 - A Q is more sensitive, because its liquid expands more.
 - B P has a larger range than Q.
 - C P is more sensitive: its liquid column moves further for each °C.
 - D They have the same sensitivity, because the liquid is the same.
6. A steel rod 2.0 m long expands by 2.0 mm when its temperature rises by 40 °C. A second steel rod, 4.0 m long, is heated by 20 °C. By how much does it expand?
 - A 1.0 mm
 - B 2.0 mm
 - C 4.0 mm
 - D 0.50 mm
7. A metal plate has a circular hole in it. The plate is heated uniformly. What happens to the diameter of the hole?
 - A It decreases, because the metal around it expands inwards.
 - B It stays the same, because the hole is empty.
 - C It increases only if the plate is thin.
 - D It increases, as if the hole were made of the same metal as the plate.
8. When a solid cube is heated, each edge becomes 0.10 % longer. Approximately by what percentage does its volume increase?
 - A 0.30 %
 - B 0.10 %
 - C 0.20 %
 - D 0.0010 %

II · Alternative to practical

[14]

A group of students builds a thermometer from a thermistor and an Arduino board. The board gives a reading **N**, a number from 0 to 1023 with no unit (counts). The students calibrate the thermometer in four water baths. In each bath they measure the temperature **T** with a reference liquid-in-glass thermometer with 1 °C divisions, and they take five readings of **N**.

Bath	T / °C	N ₁	N ₂	N ₃	N ₄	N ₅	mean N
1 (ice water)	1.0	230	226	226	226	227	227.0
2 (cold water)	13.5	309	309	311	271	312	
3 (room temperature)	24.0	391	394	394	393	393	393.0
4 (hot water)	41.5	545	547	547	543	544	545.2

(a) State the independent variable and the dependent variable of this experiment.

[2]

(b) One of the twenty readings in the table is anomalous. Circle it in the table and give one reason for your choice.

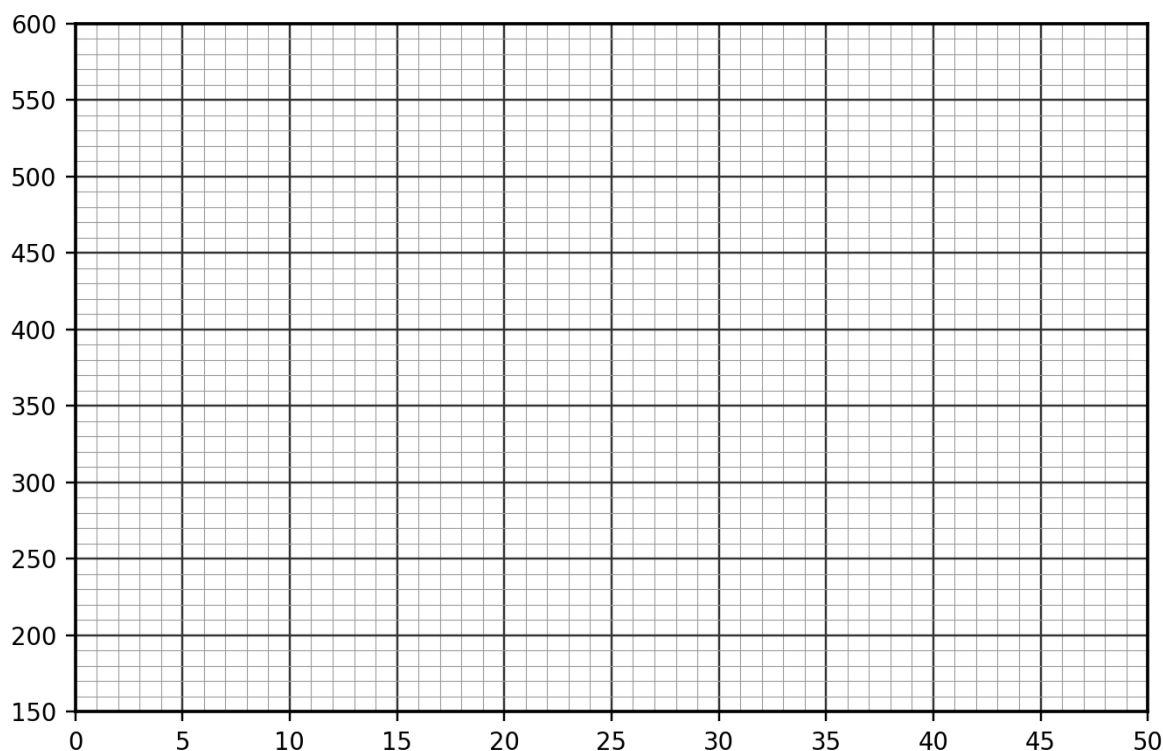
[1]

(c) The mean N for one bath has been left blank. Calculate it, ignoring the anomalous reading, and write it in the table.

[1]

(d) On the grid, plot the four mean values of N against T. The scales are printed. Mark each point with a small cross. Label both axes with the quantity and the unit.

[3]



(e) Draw a single straight line of best fit with a ruler. It should pass as close as possible to all the points, with about as many points above it as below it. Do not join the points.

[1]

(f) Use your line to find the gradient m (in counts per $^{\circ}\text{C}$). Show the triangle you use on the graph. [2]

(g) Read from your graph the value N_0 where the line crosses $T = 0^{\circ}\text{C}$. Then write the calibration equation in the form $T = (N - N_0) / m$ with your values. [1]

(h) The students now place the thermometer in a fifth bath, independent of the first four, and read $N = 483$. Use your equation to calculate the temperature of this bath. [1]

(i) The reference thermometer reads 34.6°C in the fifth bath. The client requires the Arduino thermometer to agree with the reference within $\pm 1^{\circ}\text{C}$. Does it meet the requirement? Show your calculation. [1]

(j) Suggest one specific improvement to the students' method that would make the calibration more reliable. [1]

III · Structured questions [14]

1. Temperature scales and heat

(a) Convert (i) 37°C into kelvin and (ii) 90 K into degrees Celsius. [2]

(b) A student says: "The metal spoon has a lot of heat, which is why it feels hot." Explain what is wrong with this statement. Use the words *temperature*, *heat* and *energy*. [2]

(c) Explain why a temperature of -10 K cannot exist. [1]

2. The thermometer

(a) State the two fixed points used to calibrate a Celsius thermometer, with their values. [2]

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- (b) A liquid-in-glass thermometer, taken from a cold drawer, is plunged into hot water. In the first second the liquid column drops slightly before it starts to rise. Explain why. [2]
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3. Thermal expansion

Linear expansion coefficients, α ($\times 10^{-6} \text{ }^\circ\text{C}^{-1}$):

Material	steel	copper	brass	aluminium	glass
α	12	17	19	23	9

- (a) An aluminium bar is 2.5 m long at 15 $^\circ\text{C}$. Calculate its increase in length when it is heated to 65 $^\circ\text{C}$. [3]

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- (b) A plate of the same aluminium has an area of 0.80 m² at 15 $^\circ\text{C}$. Calculate its increase in area when it is heated to 65 $^\circ\text{C}$. Use $\beta = 2\alpha$ for the area expansion coefficient. [2]
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Calettamento a caldo di un anello. In un'officina che costruisce riduttori per macchine agricole si deve montare un anello d'acciaio (massa 1,8 kg, altezza 40 mm) su un albero d'ottone. A 20 °C il foro dell'anello ha diametro 50,00 mm e l'albero ha diametro 50,06 mm: l'albero è più largo del foro e non può entrare. L'anello viene scaldato in un forno a induzione da 3,5 kW, che in circa 4 minuti lo porta alla temperatura voluta; l'albero resta a 20 °C. Per infilarlo senza forzare serve un gioco di almeno 0,04 mm, cioè il foro deve superare di 0,04 mm il diametro dell'albero. Il costruttore avverte che l'acciaio dell'anello perde la sua durezza se supera i 450 °C. Si montano 120 anelli all'ora e, in esercizio, l'albero ruota a 1500 giri/min.

Dati: coefficiente di dilatazione lineare dell'acciaio $12 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$; dell'ottone $19 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$.

Chiamiamo *interferenza* la differenza (diametro dell'albero – diametro del foro) quando i due pezzi sono alla stessa temperatura: se è positiva, l'albero è più largo del foro e il collegamento è serrato.

(a) Scaldando l'anello, il diametro del foro aumenta, diminuisce o resta uguale? Spiega in una riga. [1]

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(b) A quale temperatura minima va portato l'anello perché scivoli sull'albero con il gioco richiesto? Verifica che il valore sia compatibile con il limite indicato dal costruttore. [3]

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(c) Di quanto aumenta, in percentuale, l'area della sezione del foro alla temperatura trovata in (b)? [2]

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(d) In esercizio l'anello e l'albero arrivano insieme a 120 °C. Calcola l'interferenza a quella temperatura e dì se il collegamento resta serrato, spiegando perché. [2]

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Fine della prova.